AI EXPO 2026
at Blida University

ARTIFICIAL INTELLIGENCE EXPOSITION · APRIL 2026

Ramy Sentiment Intelligence — Understanding Customer Voice with AI

1 What We Built

Ramy Sentiment Intelligence is an AI-powered system that automatically reads and understands what customers say about Ramy products. It classifies every review into one of five categories:

Positive · Negative · Neutral · Suggestion · Question

Built specifically for the way Algerians actually write — mixing **Arabic**, **Darija**, and **French** in a single sentence — it achieves **95.06% accuracy** with full transparency into *why* each decision was made.

2 The Challenge

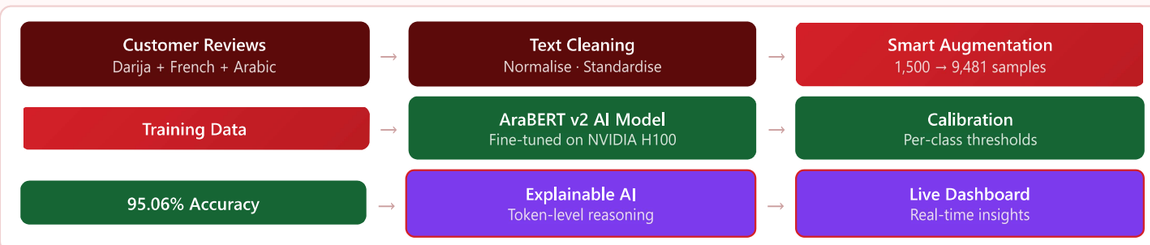
Why is this hard? Traditional AI tools fail on Algerian text because customers mix three languages freely. Imagine a review like: "*عصير Ramy بنين but الكارتون mauvais*" — no off-the-shelf tool handles this reliably.

Our goals:

- Build a **5-category classifier** that truly works on real Algerian consumer voices
- Guarantee **honest metrics** — no inflated scores from data leakage
- Show **why** the AI made each decision (Explainable AI)
- Deliver results through a **live, real-time dashboard**

3 How It Works

End-to-End Pipeline:



Key technical choices:

- **AraBERT v2** — pre-trained on Maghrebi Arabic, ideal for Darija
- **Teacher-Student learning** — the model teaches itself from unlabeled data
- **Leakage-free evaluation** — test data never seen during training or tuning

4 Performance Results

Technology: PyTorch 2.x · Transformers · scikit-learn · FastAPI · Streamlit

95.06% Accuracy	**94.37%** Macro-F1 Score
5 Sentiment Classes	**H100** NVIDIA GPU

Detailed breakdown (held-out test set):

Category	Precision	Recall	F1	Samples
Suggestion	1.00	0.85	0.92	13
Negative	0.94	1.00	0.97	16
Neutral	0.92	0.86	0.89	14
Positive	0.96	1.00	0.98	25
Question	0.93	1.00	0.96	13

Confusion Matrix:

	Sug	Neg	Neu	Pos	Que
Sug	**11**	0	1	1	0
Neg	0	**16**	0	0	0
Neu	0	1	**12**	1	0
Pos	0	0	0	**25**	0
Que	0	0	0	0	**13**

Rows = actual category · Columns = AI prediction · All metrics on strictly held-out test data

5 What Makes This Special

- **100% recall on Positive & Questions** — never misses a satisfied customer or a purchase inquiry
- **Code-switching mastery** — handles "برافو bon" and "mauvais جدا" seamlessly
- **Honest scores** — 3 leakage safeguards ensure real-world reliability
- **From 1,500 to 9,481 samples** — smart augmentation without hallucinated data

💡 **Explainable AI (XAI) — Key Innovation**

Our system doesn't just predict — it **explains why**. Using **Attention Rollout** and **Gradient×Attention**, it highlights which words drove each decision. For every prediction, stakeholders can see a color-coded evidence map showing the model's reasoning in real time.

📈 **Business Value for Ramy**

Actionable insights: Automatically detect product complaints, surface improvement suggestions, and identify purchase-intent questions across thousands of reviews. Enable data-driven decisions for product teams, marketing, and customer service — all in real time, at scale.

6 Conclusion & Impact

- **95% accuracy** on 5-class Arabic sentiment — from just 1,500 labeled samples
- **Trustworthy metrics** via leakage-free evaluation protocol
- **Transparent AI** with token-level explanations for enterprise trust
- **Next steps:** Multi-turn dialogue analysis, real-time social media monitoring, and a production API for Ramy's customer intelligence platform

Samir Guenchi — ENSIA
Senouci Amira — ENSIA
Abdennour Guerroudj — USTHB
Megamez Abderrahmane — USTHB